Independent disaster-plan analysis

This is an independent analysis, not government approval or an operational order.

Report: evaluated

Source: Quezon City DRRMP 2021-2027 Volume 1
SHA-256: 95b5c5e1177946c45f71412599df914de365feb6f6694c684d9fabaefde4990b

Modeled scope: Talayan-centered QC pilot cell; exact model-cell polygon ((121.005, 14.629), (121.018, 14.629), (121.018, 14.638), (121.005, 14.638), (121.005, 14.629)); one hypothetical 100-person cohort; no full jurisdiction estimate

Exclusions

- Rest of Quezon City: Outside this bounded pilot cell and cohort
- Only the named 100-person hypothetical cohort is modeled; no plan-wide population is allocated
- Neighboring facilities and populations are excluded; their availability and permission are unknown
- The rectangle is a modeling cell, not an administrative barangay boundary
- Road-node proximity is checked; physical connector access and shelter operability are assumptions

Source and supplemental evidence

Field	Source value	Unit	Citation
city_population_context	2.86 million	PERSON	Original PDF physical page 36
flood_priority_barangays	Talayan; Damayang Lagi; Roxas; St. Peter; Dona Imelda	TEXT	Original PDF physical page 39
shelter_capacity_people	1	COUNT	Original PDF physical page 82
opening_water_ml	unknown	ML	Original PDF physical page 84
qc_population_context	The plan estimates a city population of 2.86 million in a section citing the 2010 census; this is historical city context, not selected-area evacuation demand.	source text	Original PDF physical page 36
qc_flood_priority	Talayan, Damayang Lagi, Roxas, St. Peter and Dona Imelda are listed as the top flood-impact priority barangays. This ranking does not define pickup demand or road closures.	source text	Original PDF physical page 39
qc_district_shelter_target	One permanent evacuation area per each of six districts is a 2021-2027 project output, not six verified current shelters or capacities.	source text	Original PDF physical page 82
qc_portable_water_target	Portable drinking-water stations are mentioned with a purchase activity; no usable initial shelter water volume is stated.	source text	Original PDF physical page 84
qc_water_stock_gap	Water purification tablets and safe-drinking-water objectives are listed; initial per-shelter potable stock and delivery rate remain unknown.	source text	Original PDF physical page 87
qc_annex_gap	Page 111 lists Annexes A-D, while the supplied volume ends with a blank page 112; the substantive annex contents are not present in this source.	source text	Original PDF physical page 111, 112
reviewed road graph	d03bc52f0d1337281020eeb100c7d1de289798eb52393651f14ecd70e4f8845d	SHA-256	release/validation/geography/talayan-small.json; https://download.openstreetmap.fr/extracts/asia/philippine

Readiness gaps

- evacuee_people: The document does not establish this input. Review a scoped cohort count or declare a bounded modeling assumption. Depends: evacuated population, arrival time
- shelter_capacity_people: Only proposed or target resources are documented. Resolve facility identity and FAMILY units, or declare a reviewed capacity assumption. Depends: shelter feasibility, unserved population
- vehicle_seats: The document does not establish this input. Supply current fleet seats or declare a scenario assumption. Depends: trip feasibility, arrival time
- opening_water_ml: Only proposed or target resources are documented. Supply dated stock evidence or declare a scoped water assumption. Depends: water sufficiency, water shortfall
- Missing annex: Annex A contingency plans
- Missing annex: Annex B operations center manual
- Missing annex: Annex C prior-plan evaluation
- Missing annex: Annex D capacity building
- Procedure not modeled: walking
- Procedure not modeled: staged transfers
- Procedure not modeled: resupply

Explicit assumptions

- Personally reviewed local model cell and OSM connectors; physical access and facility operation assumed
- One hypothetical 100-person cohort, 20-seat prepositioned vehicle, one dedicated crew, 150-person shelter with zero initial occupants; 80 people remain at pickup
- Opening water: pickup 100,000 mL and shelter 400,000 mL; 2,000 mL/person/day over 259,200 seconds. Hypothetical change adds 500,000 mL at pickup
- Passenger deadline 216 seconds; one second positioning, loading and unloading each
- Road closure at time zero on exact edge osm:107948780:13:forward; known to policy at time zero
- Car-road routing uses a declared 30 km/h fallback where OSM speed is unknown; the selected anchors and route do not establish field travel or accessibility
- No FAMILY-to-PERSON conversion; city/municipal historic totals and proposed resources are context only
- Variant is hypothetical and untested in the field; no resupply during the three-day horizon

Measured outcomes

Metric	Value	Unit
normal baseline on-time arrival	20	people
normal baseline unmet water	380099	mL
normal car route travel	211	seconds
road_close baseline on-time arrival	0	people
road_close baseline unmet water	380103	mL

Metric	Value	Unit
road_close car route travel	221	seconds
normal added-water on-time arrival	20	people
normal added-water unmet water	0	mL
normal car route travel	211	seconds
road_close added-water on-time arrival	0	people
road_close added-water unmet water	0	mL
road_close car route travel	221	seconds

Regressions

- No regression under the recorded four-metric paired comparison

Attribution

- Independent exploratory analysis; hypothetical assumptions are not plan facts
- Road geometry © OpenStreetMap contributors, ODbL 1.0; regional snapshot 2026-10-13
- No field predictive validation or government approval is claimed